

IntelliDesk AI

AI-Powered Email Support Automation for B2B SaaS

Designed to classify, prioritize, deduplicate, and resolve support tickets in real time.

Agglomeration 2.0 | Team CofeeSeekers
Built with Next.js, n8n, Gemini, Pinecone, Supabase

The Support Dilemma: Why IntelliDesk Matters

The Scale Problem

200–500 emails/day
Multiple categories (tech, billing, access, how-to)
30–45 min manual effort per email
Repetitive queries waste agent time

Operational Risks

Urgent tickets buried in noise
SLA breaches become common
Escalations damage trust
No intelligent prioritization

Baseline First Response Time

2-3 Hours

Baseline CSAT Score

6.5 / 10

Core Requirements: Building the Intelligent Support Solution

Our development was guided by key requirements to create a robust and reliable AI-powered support system.

“
AI-Based Context Classification
Classifies emails into 8+ categories using Gemini → outputs structured JSON with confidence score.
”

“
Thread Detection & Deduplication
Combines thread ID matching (Supabase) + semantic similarity (>0.85 via Pinecone) to prevent duplicates.
”

“
Urgency & Severity Detection
Assigns P1–P4 using intent, keywords, tone signals, and escalation logic.
”

“
Intelligent Auto-Response
Auto-replies only when confidence ≥ 0.80 and severity $\neq$ P1 (safety gate enforced).
”

“
Customer Identification
Maps sender domain → identifies tier, account context, and routing priority.
”

“
Structured Ticket & Dashboard
Stores metadata (priority, tone, confidence) for full visibility and auditability
”

Context-Aware Analysis

Our system extracts sentiment, intent, and technical signals like error codes (e.g., PG-5023) from incoming emails.

RAG with Pinecone

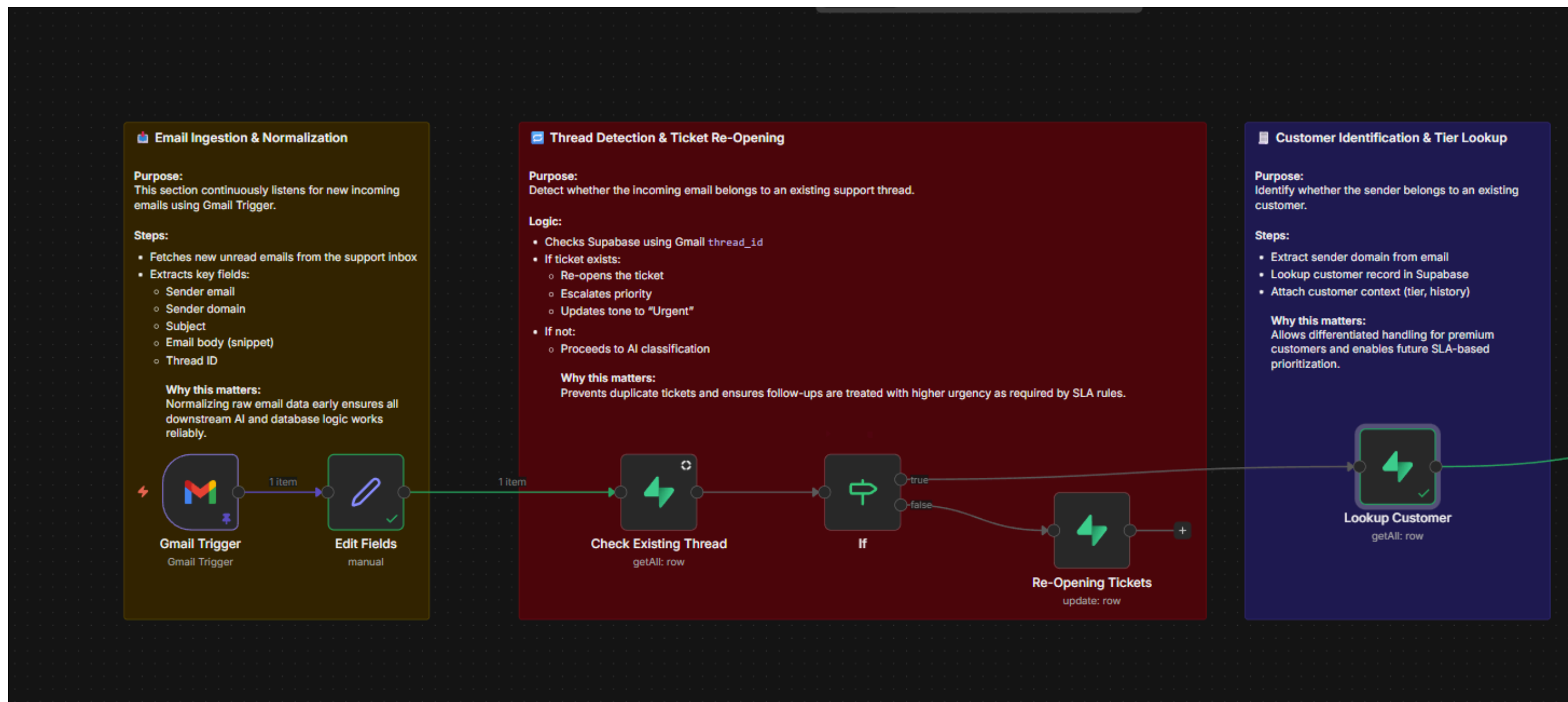
Retrieval-Augmented Generation (RAG) uses Pinecone to ground responses in historical knowledge. Similarity scores >90% indicate a Perfect Match, leading to highly accurate and relevant answers.

Customer 360 View

Agents have a complete view of the customer, including account context, service tier, and full ticket history, all within the same workspace.

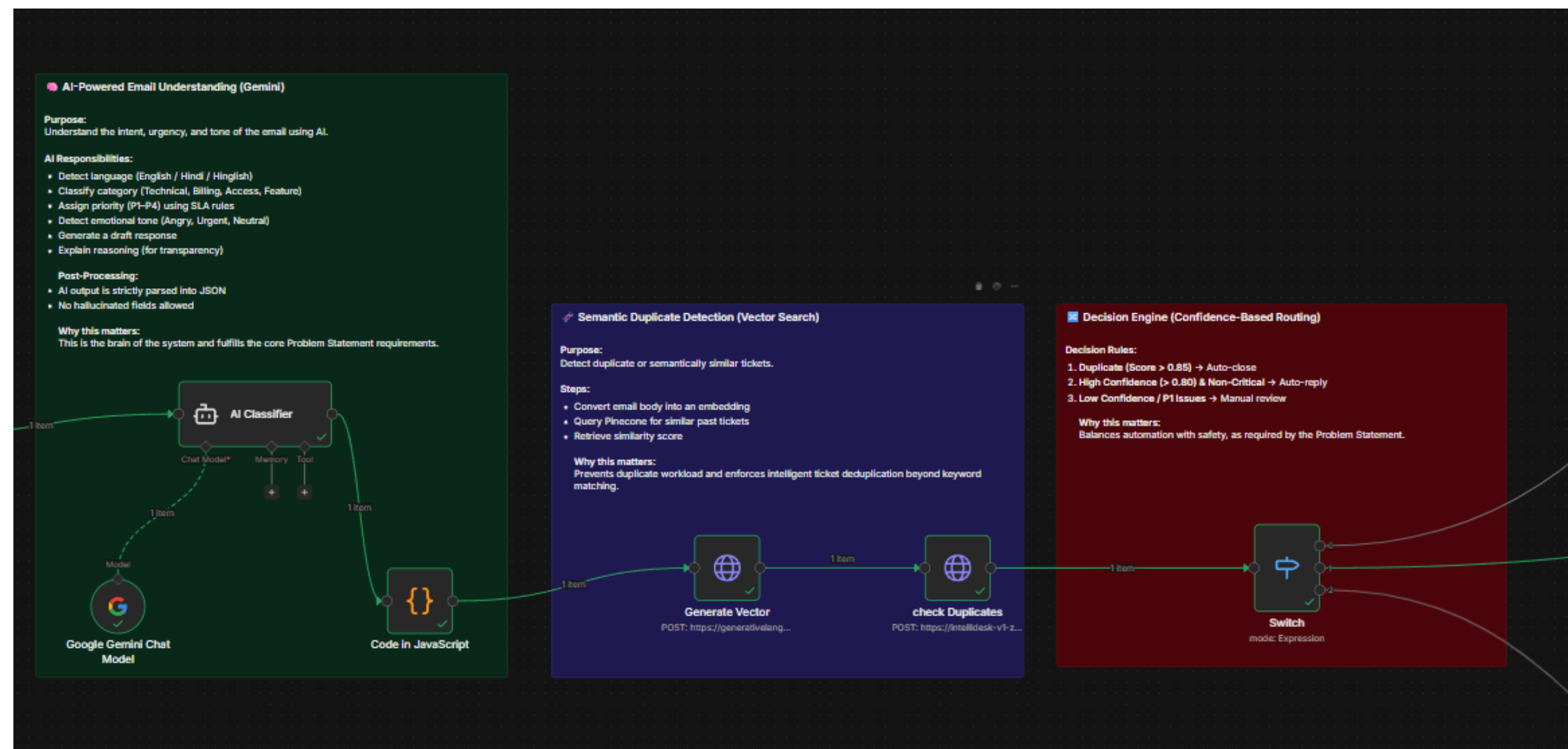
Resource Integration

IntelliDesk intelligently suggests relevant FAQs, documentation, and guides based on the semantic meaning of the query, empowering faster resolutions.



Intelligence Pipeline:

Classification → Deduplication → Decision Engine

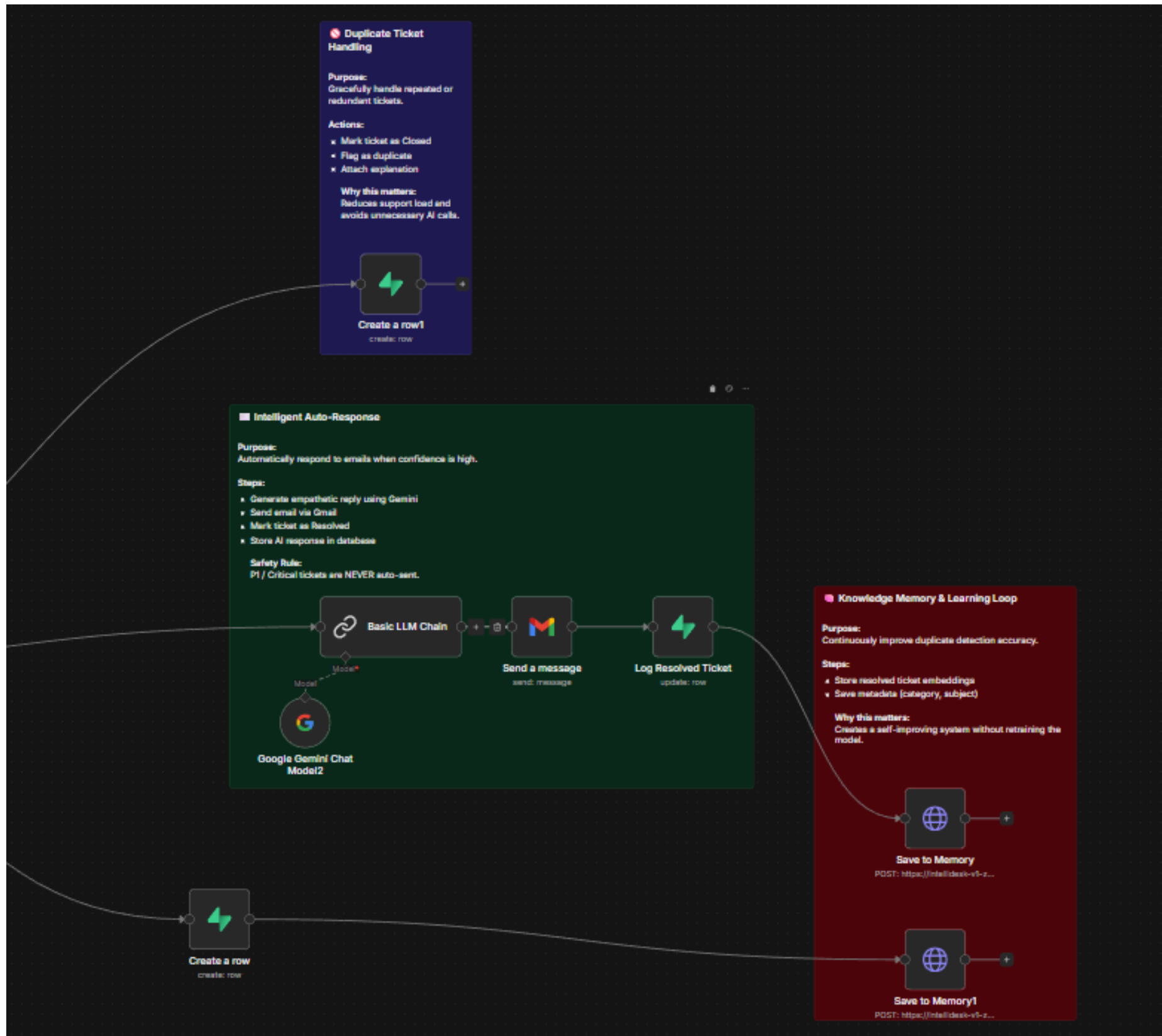


Core Backend Logic (n8n Workflow)

- Gemini Classifier
- Outputs structured
- JSON → category, priority, tone, confidence
- Vector Deduplication (Pinecone)
- Embeddings similarity > 0.85 → duplicate detection
- Decision Engine (Safety Layer)
 - Duplicate → Auto-close
 - Confidence ≥ 0.80 and Non-P1 → Auto-reply
 - P1 or Low confidence → Human queue

Design Principle:
Automate when safe. Escalate when critical.

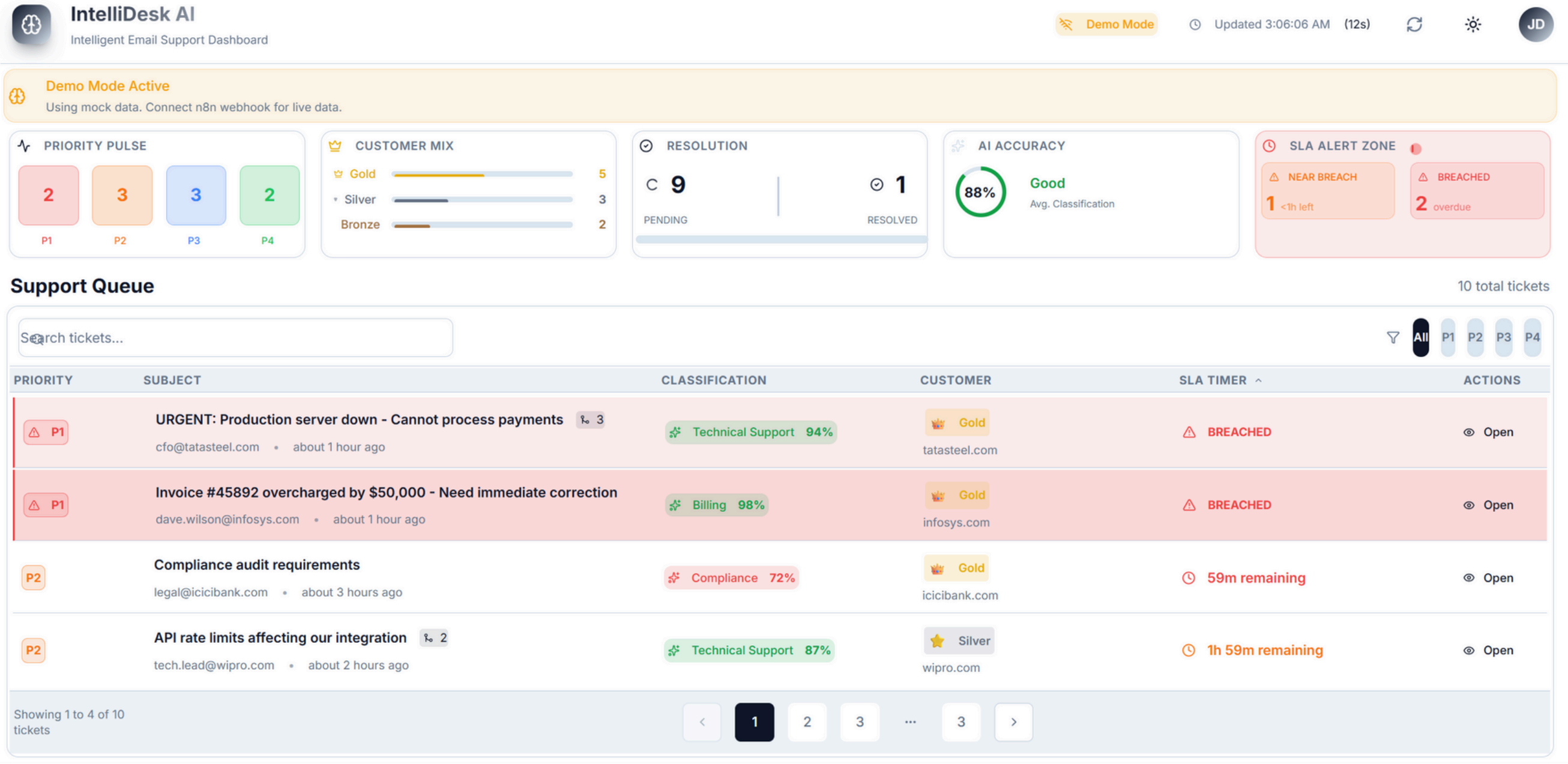
Execution Paths & Learning Loop (Post-Decision Engine)



System Behavior After Decision

- **Duplicate Ticket Handling**
 - Mark ticket closed
 - Attach explanation
 - Prevent unnecessary agent workload
- **Intelligent Auto-Response (Safe Mode)**
 - Gemini drafts response
 - Sent only when confidence ≥ 0.80 and not P1
 - Ticket auto-marked resolved
 - Response stored for traceability
- **Knowledge Memory & Learning Loop**
 - Stores embeddings of resolved tickets
 - Improves future duplicate detection
 - System improves without retraining model

Real-Time Intelligence Dashboard for Support Operations



What this dashboard enables

- **Priority Pulse (P1–P4)**
 - AI assigns severity using intent, urgency, and business impact.
- **Customer Tier Context**
 - Domain-based mapping → Gold / Silver / Bronze visibility.
- **AI Transparency**
 - Every prediction shows confidence (e.g., 94%, 98%).
- **SLA Intelligence**
 - Live timers flag near-breach and breached tickets.

Operational Outcome:

Teams focus on critical work instead of scanning inbox noise.

AI-Assisted Resolution Workspace (RAG in Action)

Back to Queue

TKT-002

Customer

P1

SLA BREACHED

Invoice #45892 overcharged by \$50,000 - Need immediate correction

Dave Wilson

<dave.wilson@infosys.com>

Jan 25, 2026 1:58 AM

infosys.com

Hi Billing Team,

I just received Invoice #45892 dated Jan 20, 2024 and the amount is completely wrong.

Invoice shows: \$75,000
Our contract rate: \$25,000/month

That's a \$50,000 overcharge! I've attached our signed contract for reference (see MSA-2023-INF-0234.pdf).

This needs to be corrected immediately as our finance team has flagged this for audit. We cannot process payment until this is resolved.

Please issue a corrected invoice within 24 hours or we'll need to escalate to procurement leadership.

Thanks,
Dave Wilson
Senior Procurement Manager
Infosys Technologies
Mumbai, India

AI Response

Perfect Match 98%

Billing

Status: In Progress

Sentiment: Frustrated

Dear Dave,

Thank you for bringing this to our attention. I have reviewed Invoice #45892 and confirmed the discrepancy you mentioned.

Investigation Results:

- Your contracted rate: \$25,000/month ✓
- Billed amount: \$75,000 (incorrect)
- Variance: \$50,000 overcharge

This error occurred due to an incorrect billing code being applied during our system migration last week.

Resolution:

1. Credit memo for \$50,000 has been issued (CM-2024-0123)
2. Corrected invoice will be sent within 2 hours
3. This will reflect in your account within 24-48 hours

I sincerely apologize for this inconvenience. Please let me know if you need anything else.

Best regards,
IntelliDesk AI Billing Support

Edit Response

Copy

Send Response

Related Resources

Invoice Correction Process

Credit Memo FAQ

Infosys Technologies

infosys.com

Gold

ACCOUNT OVERVIEW

Lifetime Value

\$890,000

Past Tickets

45

Last Contact

12 days ago

Account Manager

Rahul Kapoor

Give support agents a grounded, context-rich response — not a generic AI reply.

What this screen demonstrates:

- Context Understanding:
- The system identifies Billing issue, P1 severity, and frustrated sentiment automatically.
- RAG-Backed Response (98% match):
- Draft is generated using similar historical tickets and internal knowledge — not hallucination.
- Explainable AI Output:
- Structured reasoning: investigation results, resolution steps, and timeline are visible and editable.
- Human-in-the-Loop Control:
- Agent can edit, review, or reject the response before sending.
- Customer 360 Intelligence:
- Account tier (Gold), lifetime value, ticket history, and account manager are visible in the same view.